

Recap on change of variable for integration

April 17, 2026

def: Let D^* and D be subsets of $\mathbb{R}^n$. A coordinate transformation (from D^* -coordinates to D -coordinates) is a function $T: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that:

- 1) $T(D^*) = D$ and the map $T: D^* \rightarrow D$ is bijective, #
i.e. one-to-one (injective) and onto (surjective)
- 2) T is C^1 (partial derivatives exist and are continuous)
- 3) $\det(DT) \neq 0$ on all D^*

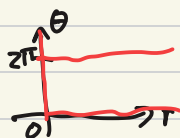
notation: Typically, the (original) D -coordinates are Cartesian $(x, y, z, \dots)$ and the (new) D^* -coordinates are $u, v, w, \dots$ or $r, \theta, \phi, \dots$

The Jacobian matrix is $DT = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \dots \\ \frac{\partial y}{\partial u} & \dots & \ddots \\ \vdots & \vdots & \ddots \end{bmatrix}$.

The Jacobian is the determinant of the Jacobian matrix and is denoted $\det(DT)$ or $\frac{\partial(x, y, \dots)}{\partial(u, v, \dots)} = \det(DT)$
(here $x, y, \dots$ are the original coordinates, and $u, v, w, \dots$ are the new coord.)

remark: We sometimes allow a change of variable that isn't injective and/or has $\det(DT) = 0$ at some point.
We can only do that if the set where injectivity or $\det(DT) \neq 0$ fails has no length/area/volume:

e.g. Polar coordinates $T(r, \theta) = (r \cos \theta, r \sin \theta) : [0, \infty) \times [0, 2\pi] \rightarrow \mathbb{R}^2$
has $\det(DT) = 0$ only along the line $r = 0$
and is not injective only along $r = 0$ and on $\theta = 0, 2\pi$
This doesn't affect area integrals because these lines have 0 area.



Change of variables in 1-D: This is just an application of the Fundamental Theorem of Calculus and Chain Rule

$$\begin{aligned}\int_a^b f(x) dx &= F(b) - F(a) = F(g(b^*)) - F(g(a^*)) \\ &= \int_{g(a^*)}^{g(b^*)} f(x) dx = \int_{a^*}^{b^*} f(g(u)) \frac{dx}{du} du\end{aligned}$$

Change of variables in double integrals:

Given a coordinate transformation T from D^* to D , and an integrable function f on D , then

$$\iint_D f(x, y) dx dy = \iint_{D^*} f(x(u, v), y(u, v)) |\det(DT)| du dv$$

notation: $\iint_D dx dy$ is a double integral, the iterated integral might be in the order $dx dy$ or $dy dx$

same for $\iint_{D^*} du dv$.

Change of variables in triple integrals:

Given a coordinate transformation T from D^* to D , and an integrable function f on D , then

$$\iiint_D f(x, y, z) dx dy dz = \iiint_{D^*} f(x(u, v, w), y(u, v, w), z(u, v, w)) |\det(DT)| du dv dw$$

notation: $\iiint_D dx dy dz$ is a triple integral, the iterated integral might be in the order $dx dy dz$ or $dy dx dz$, or any of the other orders

same for $\iiint_{D^*} du dv dw$.

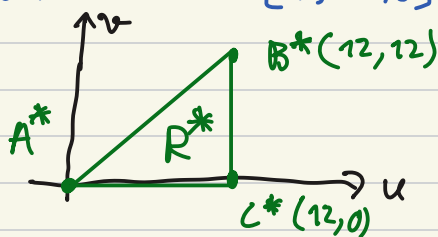
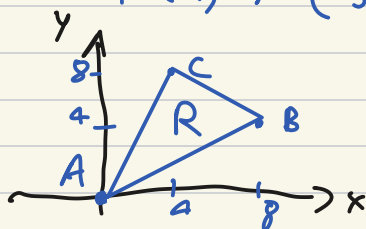
Linear transformation: $T(u, v) = M \begin{pmatrix} u \\ v \end{pmatrix}$
 where M is a 2×2 matrix with $\det(M) \neq 0$

e.g. $\iint_R \sin(x+y) \cos(2x-y) \, dx \, dy$

where R is the triangle $A(0,0), B(8,4), C(4,8)$
 change of variable $u = x+y \quad v = 2x-y$

$$T(u, v) = \left(\frac{u+v}{3}, \frac{2u-v}{3} \right)$$

$$M = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & -1/3 \end{bmatrix}$$



$$DT = M$$

$$\frac{\partial(x,y)}{\partial(u,v)} = \det M = -\frac{1}{3}$$

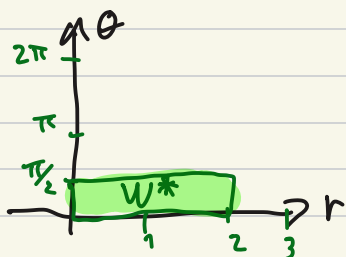
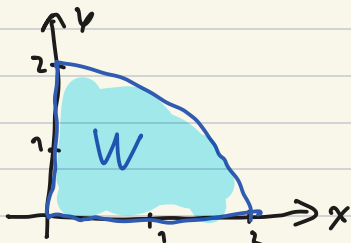
$$\begin{aligned} \iint_R \sin(x+y) \cos(2x-y) \, dx \, dy &= \int_0^{12} \int_0^u \sin(u) \cos(v) \left| \frac{-1}{3} \right| \, dv \, du \\ &= \frac{1}{3} \int_0^{12} \sin(u) \left[\int_0^u \cos v \, dv \right] \, du \\ &= \frac{1}{3} \int_0^{12} \sin(u) \left[\sin v \Big|_{v=0}^{v=u} \right] \, du \\ &= \frac{1}{3} \int_0^{12} \sin^2(u) \, du = \frac{1}{3} \int_0^{12} \frac{1 - \cos(2u)}{2} \, du \\ &= \frac{1}{3} \left(\frac{u}{2} - \frac{\sin(2u)}{4} \Big|_{u=0}^{u=12} \right) = \frac{12}{6} - \frac{\sin(24)}{12} \end{aligned}$$

Polar coordinates in 2-D

$$T(r, \theta) = (r \cos \theta, r \sin \theta)$$

e.g. $\int_0^2 \int_0^{\sqrt{4-y^2}} \sqrt{x^2+y^2} dx dy$

$$W = \{ (x, y) \in \mathbb{R}^2 : 0 \leq y \leq 2, 0 \leq x \leq \sqrt{4-y^2} \}$$



$$DT = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}$$

$$\begin{aligned} * \sqrt{x^2+y^2} &= \sqrt{r^2 \cos^2 \theta + r^2 \sin^2 \theta} \\ &= \sqrt{r^2} \end{aligned}$$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \det(DT) = r \cos^2 \theta + r \sin^2 \theta = r$$

$$\int_0^2 \int_0^{\sqrt{4-y^2}} \sqrt{x^2+y^2} dx dy = \int_0^2 \int_0^{\pi/2} \sqrt{r^2} |r| d\theta dr$$

$$= \int_0^2 \int_0^{\pi/2} r \cdot r d\theta dr$$

$$= \left(\frac{\pi}{2} - 0\right) \int_0^2 r^2 dr = \frac{\pi}{2} \frac{r^3}{3} \Big|_{r=0}^{r=2}$$

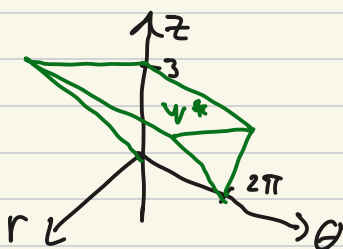
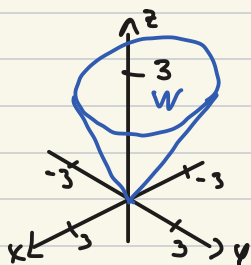
$$= \frac{\pi}{2} \frac{8}{3} = \frac{4\pi}{3}$$

Area element in 2-D polar coord. is

$$\boxed{r dr d\theta}$$

Cylindrical coord. in 3D $T(r, \theta, z) = \begin{pmatrix} r \cos \theta \\ r \sin \theta \\ z \end{pmatrix}$

e.g. $\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{\sqrt{x^2+y^2}}^3 \frac{e^z}{\sqrt{x^2+y^2}} dz dy dx$



$$W^* = \{(r, \theta, z) : 0 \leq \theta \leq 2\pi, 0 \leq r \leq 3, r \leq z \leq 3\}$$

$$DT = \begin{bmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r$$

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{\sqrt{x^2+y^2}}^3 \frac{e^z}{\sqrt{x^2+y^2}} dz dy dx$$

$$= \int_0^{2\pi} \int_0^3 \int_r^3 \frac{e^z}{r} |r| dz dr d\theta = \int_0^{2\pi} \int_0^3 \int_r^3 \frac{e^z}{r} r dz dr d\theta$$

Volume element in 3-D cylindrical coord. is $\boxed{r dr d\theta dz}$

Spherical coordinates in 3-D:

$$T(r, \varphi, \theta) = \begin{pmatrix} r \sin \varphi \cos \theta \\ r \sin \varphi \sin \theta \\ r \cos \varphi \end{pmatrix}$$

$$DT = \begin{bmatrix} \sin \varphi \cos \theta & r \cos \varphi \cos \theta & -r \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & r \cos \varphi \sin \theta & r \sin \varphi \cos \theta \\ \cos \varphi & -r \sin \varphi & 0 \end{bmatrix}$$

$$\begin{aligned} \frac{\partial(x, y, z)}{\partial(r, \varphi, \theta)} &= (-1)^{3+1} \cos \varphi \det \begin{bmatrix} r \cos \varphi \cos \theta & -r \sin \varphi \sin \theta \\ r \cos \varphi \sin \theta & r \sin \varphi \cos \theta \end{bmatrix} \\ &\quad + (-1)^{3+2} (r \sin \varphi) \det \begin{bmatrix} \sin \varphi \cos \theta & -r \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & r \sin \varphi \cos \theta \end{bmatrix} \\ &= \cos \varphi [r^2 \cos \varphi \sin \varphi \cos^2 \theta + r^2 \cos \varphi \sin \varphi \sin^2 \theta] \\ &\quad - (-r \sin \varphi) (r \sin^2 \varphi \cos^2 \theta + r \sin^2 \varphi \sin^2 \theta) \\ &= r^2 \cos^2 \varphi \sin \varphi (\cos^2 \theta + \sin^2 \theta) + r^1 \sin^3 \varphi (\cos^2 \theta + \sin^2 \theta) \\ &= r^2 \sin \varphi (\cos^2 \varphi + \sin^2 \varphi) = r^2 \sin \varphi \end{aligned}$$

Volume element in 3-D
spherical coord. is

using $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = T(r, \varphi, \theta) = \begin{pmatrix} r \sin \varphi \cos \theta \\ r \sin \varphi \sin \theta \\ r \cos \varphi \end{pmatrix}$

$$r^2 \sin \varphi \, dr \, d\varphi \, d\theta$$

$$0 \leq r$$

$$0 \leq \varphi \leq \pi$$

$$0 \leq \theta \leq 2\pi$$